

Fruit DNA: Extract, Observe, Evaluate

I can explain the steps in fruit-DNA extraction, record an observation and evaluate the evidence cautiously.

METHOD

release, filter, precipitate

OBSERVE

record, do not overclaim

CONNECT

DNA carries genes

[Open the original interactive lesson](#)

What success looks like

- I can order the extraction method and explain the role of detergent, salt, filtration and cold ethanol.
- I can record the precipitate observation without calling it pure DNA.
- I can connect extracted DNA to chromosomes and genes and state a method limitation.

Retrieve and reconnect

- Describe complementary base pairing.
- State the chromosome-DNA-gene relationship.
- Explain what filtration separates.
- Lead-in: why is observation different from identification?

**White stringy material appears where the fruit filtrate meets cold ethanol.
What is the safest scientific claim?**

Think first. Point, say, sign or write your choice. Give one reason.

- A. The material is pure DNA and nothing else.
- B. A DNA-containing precipitate formed; the appearance alone does not show purity.
- C. Fruit cannot contain DNA because it is not an animal.

Look closely · what is the evidence?

RELEASE · disrupt membranes

SEPARATE · filter solids

PRECIPITATE · DNA-containing material appears in cold ethanol

Look closely · what is the evidence?



Release



Filter



Precipitate

Words we will use

- filtrate — liquid through the filter
- precipitate — material appearing from solution
- extract — remove from a mixture

Watch the model

- Mash the fruit with salt-detergent solution: detergent disrupts cell and nuclear membranes, while salt helps DNA strands clump together.
- Filter the mixture; collect the filtrate while larger fruit solids remain behind.

Why is detergent used in this extraction?

- A. To disrupt cell and nuclear membranes so DNA can be released.
- B. To colour each DNA base.
- C. To copy genes by mitosis in the tube.

Commit to an answer. Show the evidence you used. Listen, then revise if needed.

Build the explanation

- Observation: describe colour, location and texture without writing 'pure DNA'.
- Inference: the extraction method is expected to produce DNA-containing material, but appearance alone cannot identify every component.
- Connection: the extracted DNA came from fruit-cell chromosomes and contains many genes; the extraction does not reveal individual genes or base sequence.

Find and repair the misconception

The white material proves that a pure gene has been seen.

A. Keep it: a gene is visible as one white strand.

B. Repair it: the precipitate contains DNA, may contain other material and does not reveal individual genes.

C. Repair it: only animal cells contain DNA.

Observation, inference or limitation?

Classify each statement after the practical or parity dataset, then repair any overclaim.

- METHOD
- OBSERVE
- INFER
- LIMIT
- CONNECT

Observation, inference or limitation?

Use the numbered cards in your pupil pack. Take turns to choose, justify and check. You may direct a partner to move a card.

Choose a group · defend your placement

- DIRECT OBSERVATION
- SCIENTIFIC INFERENCE
- METHOD LIMITATION

Say what was observed, what was inferred and what cannot be concluded.

Your independent mission

Carry out or direct the non-core fruit-DNA extraction, record the result and evaluate the evidence.

Record: An ordered method, precise observation, DNA/chromosome/gene connection and one valid evaluation point.

Choose your support and challenge

- Supported: Order four method pictures, select the observed result and complete one limitation sentence.
- Standard: Record method and observation, explain two reagent/separation roles and evaluate one limitation.
- Stretch: Compare practical/parity outcomes, identify variables that affect visible yield and propose one fair improvement.

Show what you know

Why is 'DNA-containing precipitate' better than 'pure DNA'?

Point to, read or explain the evidence you made. Choose one next step after the adult has listened.